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① Compare Type-1 and Type-2 Hypervisors

Introduction

A hypervisor is software that creates and manages virtual machines (vms). there are two types of hypervisors. Type-1 (Bare-metal) and Type-2 (Hosted).

Comparison

Feature	Type-1 Hypervisor	Type-2 Hypervisor
installation	Runs directly on hardware	Runs on a host OS
Performance	High	Moderate
security	strong	Lower
Examples	vmware ESXi, HyperV	virtualBox, VMare workstation

Analysis:

- performance: Type-1 provides better CPU and memory performance because it runs directly on the none other than hardware.

• **Security:** It has a smaller attack surface and offers better VM isolation than Type-2.

• **Manageability:** supports centralized management, live migration and load balancing, and high availability, making it suitable for enterprise cloud environments.

Justification:

For mission-critical applications, Type-1 Hypervisor is the better choice because it provides higher performance, stronger security, greater scalability, and improved reliability.

Conclusion:

Type-1 hypervisors are widely used in cloud data centers because they ensure efficient resource utilization and secure virtualization.

② Virtualization Architecture for workload Isolation:

Introduction

Virtualization allows multiple workloads such as Finance, Healthcare, and student portal applications to run securely on the same physical server while keeping them isolated.

How virtualization supports Isolation:

• Hypervisor Isolation: Each VM operates independently, preventing interference between workloads.

- Memory Isolation: Every VM has dedicated virtual memory, preventing unauthorized access.
- storage Isolation: separate virtual disks ensure data security.
- Network isolation: VLANs and virtual switches separate network traffic.

Major Attack Surfaces

1. Hypervisor Attacks:

- VM Escape
- Hypervisor vulnerabilities

2. virtual Network Attacks:

- Packet sniffing
- ARP spoofing
- VLAN hopping

Security measures

- Regular hypervisor updates
- Firewalls and encryption
- Network segmentation
- strong access control.

Conclusion:

virtualization ensures secure workload isolation, but hypervisor and network attacks should be controlled using proper security measures.

③ VM Host utilization Analysis:

Given Data:

- CPU utilization = 92%
- Memory usage = 88%
- storage I/O Latency = 35ms
- VM Boot Time = 170 seconds.

Likely Bottlenecks

- CPU: High utilization causes slower processing and delays
- Memory: High usage may lead to swapping and reduced VM performance.
- storage: High Latency slows data access and increases application response time.
- Boot time: Long VM startup indicates storage and CPU bottlenecks.

Corrective Actions

- upgrade CPU and memory
- migrate some VMs to another host
- Replace HDDs with SSD/NVMe storage
- optimize VM images and remove unused VMs.

Conclusion:

The host is overloaded due to CPU, memory and storage limitations. Resource upgrades and workload balancing will improve overall cloud performance.

(4) Software Denied Datacenter (SDDC)

Introduction

A software Defined Datacenter (SDDC) virtualizes compute, storage, and networking allowing resources to be managed through software instead of manual hardware configuration.

Virtualization components

Compute Virtualization

- Creates mutiple VMs.
- Enables quick provisioning and better resource utilization.

Storage Virtualization

- combines physical storage into a single storage pool.
- Simplifies backup and improves scalability.

Network virtualization

- Uses SDN, virtual switches, and virtual routers.
- provides flexible and secure network management.

Advantages over Traditional Datacenter.

- Faster deployment
- Automated resource management
- Better scalability
- Reduced operational cost.

Conclusion:

SDDC improves agility by automating compute, storage and networking, making cloud infrastructure more efficient.

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Multi - cloud Federation identify

Introduction:

multi- cloud Federation allows vers to access services from different cloud providers using a single identity. Identity mismatches occur when providers use different authentication methods.

Causes to identify Mismatch:

- Different User databases
- Different authentication protocols
- Lack of trust between cloud providers
- Inconsistent user roles and permissions

Secure identify Design:

- Uses a centralized identify provider
- Implement single sign-on
- Use federation protocols such as SAML, OAuth 2.0 and openID Connect.
- Enable multi- Factor Authentication and Role - Based Access control.
- protect data using encryption and audit logging.

Benefits:

- secure authentication
- Better privacy

Conclusion:

A secure Federation design using IdP, SSO, MFA and RABC eliminates identity mismatches and provides safe, seamless access across multiple cloud providers.